Chapter 4 Teacher Key — Energy Transfer

> Combined answer keys: Sound Transfer · Light Transfer · Heat Transfer · Electricity Transfer · Practice · Chapter 4 Quiz.

Answer Key — Sound Transfer (Chapter 4)

- **Artifact:** `production/worksheets/04-ch4-sound-transfer-student.docx`
- **Source alignment:** TG Lesson 6 (cup-and-paper demo) · AP 6.1 · Reader pp. 17, 19

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception	Sco not
Big Question	Sound transfers energy from one place to another; evidence = the paper moved without contact.	"Sound made the paper move without touching it."	sound → paper moved	Change without contact = evidence of transfer	"Sound is matter that flies to the paper."	Full for nam the cha no con
Prediction	Any reasonable prediction ("the	—	prediction stated before test	—	—	Any thou pred (not grad

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception	Sco note
	paper will move / stay still").					corr ss)
Observe (distance table)	Paper movement recorded as Clear/Slight/None per distance; evidence column names the change.	Qualitative words (moved a lot/little/not at all)	closer → clearer movement (qualitative)	—	Treating distance as exact measurement	Cred reco obse ns, l perf dista tren
Explain	The paper moved even though nothing touched it; that change is evidence sound	"Sound energy traveled through the air to the paper."	change + no contact → transfer	The change in the target is evidence	"The air pushed the paper" (overly mechanistic)	Cred infe do N requ wav cle c

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception	Sco note
	transferred energy to the paper.					

Notes

- Do NOT require a perfect near/middle/far distance trend — distance is qualitative here.
- Do NOT require "compression wave" or "vibration of air particles."
- Do NOT penalize "sound travels through air" — it's a reasonable Grade 4 statement (the TG uses "sound waves from tapping the cup pushed air").

Answer Key — Light Transfer (Chapter 4)

- **Artifact:** `production/worksheets/04-ch4-light-transfer-student.docx`
- **Source alignment:** TG Lesson 7 (flashlight demo) · AP 8.1 Station 1 (chalk dust removed) · Reader p.19

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception
Big Question	Light transfers energy from the flashlight (source) to the target; evidence =	"Light traveled from the flashlight to the card."	light reaches target → target change	The lit target is the change; transfer is	"I saw energy." (students see the change, not energy)

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception
	the target lights up.		s	inferred	
Prediction	Any reasonable prediction ("the target will light up / a shadow appears").	—	stated before test	—	—
Source /path/ target trace	SOURCE = flashlight → PATH = beam → TARGET = card → OBSERVED CHANGE = lit/bright spot.	order correct with any wording	flashlight → beam → card → lit	the model orders the transfer	naming the card as the source
Barrier / shadow test	Blocking the path removes the lit spot; removing the barrier brings it back.	"The shadow appeared when I blocked it."	blocked path → no change	change follows the beam → transfer follows the path	"Light stopped existing."
Explain	The target lit up when the beam reached it; that change	"Light energy moved from the flashlight	change in target = evidence	students observe the change,	"I directly saw energy."

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception
	is evidence light transferred energy from the flashlight to the target.	to the card."		infer transfer	

Notes

- Do NOT require chalk-dust beam visualization; the target-lighting change is sufficient evidence.
- Redirect "I saw energy" → "You saw light reach the target and the change it caused."

Answer Key — Heat Transfer (Chapter 4)

- **Artifact:** `production/worksheets/04-ch4-heat-transfer-student.docx`
- **Source alignment:** TG Lesson 7 (hot-plate demo, replaced) · AP 8.1 Station 2 (hot plate replaced with warm water) · Reader p.20

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception
Big Question	Heat transferred energy from the warmer water to the cooler object;	"The warm water got cooler and the cup got warmer."	warmer cools + cooler warms	temperature change = evidence of heat transfer	"Cold moved into the water."

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception
	evidence = the temperatures changed (one cooled, one warmed).				
Prediction	Any reasonable prediction ("the cool object will warm up").	—	stated before test	—	—
Initial temperatures	Realistic initial readings in °C (e.g., warm water ~38 °C, cooler object ~21 °C).	any plausible values with °C	warm > cooler at start	—	no units / impossible values
Later temperatures	Warm water decreased, cooler object increased (e.g., 34 °C and 27	any plausible direction	warm ↓, cooler ↑	energy moves warmer → cooler	"both got warmer" / "cold moved"

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception
	°C).				
Explain	The warmer material cooled and the cooler material warmed — the temperature change is evidence that heat energy transferred from the warmer to the cooler material.	"Heat moved from the warm water to the cool cup."	warmer → cooler	transfer direction is warmer → cooler	"heat and temperature are the same"

Notes

- Temperature units must be °C and consistent (source/project convention). No °F.
- Reject impossible values (e.g., water at 90 °C, or the cooler object hotter than the water after).
- Do NOT require conduction/convection/radiation terms.

Answer Key — Electricity Transfer (Chapter 4)

- **Artifact:** `production/worksheets/04-ch4-electricity-transfer-student.docx`

- **Source alignment:** TG Lesson 7 (electricity-generation video) · AP 8.1 Station 3 (wall outlet replaced with low-voltage battery) · Reader p.21

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Commo misconc ion
Big Question	Electric current transferred energy from the battery to the device; evidence = the device changed (lit / sounded / moved).	"The battery powered the bulb through the wires."	battery → circuit → device → change	the distant device's change = evidenc e of transfer	"Electric was use before reaching device."
Prediction	Any reasonable prediction ("the bulb will light when the circuit is complete").	—	stated before test	—	—
Model (BATTERY → CIRCUIT → DEVICE → CHANGE)	Battery = source; circuit = path; device = target; change = light/sound/ motion.	order correct with any wording	battery → circuit → device → change	the model orders the transfer	naming device a the sour
Observe (complete vs broken circuit)	Complete circuit → device	"It stops when I disconne	comple te → change	energy transfer follows	"the wire carries 'used'

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception
	changes; broken circuit → change stops.	ct a wire."	; broken → none	the circuit path	electricit
Classify TRANSFER / TRANSFORMATION / BOTH	**BOTH** — energy moved through the wires (transfer) AND changed form to light/sound/motion (transformation).	BOTH with either justification	BOTH	the device changes form at a distance	"only transfer" "only transformation")

Notes

- This is the anchor system for "BOTH." Ensure students can name BOTH the movement (transfer) and the form change (transformation).
- Do NOT require electron flow, circuit equations, or current/voltage terms.
- Safety: low-voltage battery only — the key should never reference wall outlets/chargers as the investigation.

Answer Key — Energy Transfer Practice (Chapter 4)

- **Artifact:** `production/worksheets/04-ch4-energy-transfer-practice.docx`
- **Source alignment:** Reader Ch.4 pp. 17–24 · TG Lessons 6–9 · AP 7.1 / 9.1 / 9.2

Section A — FIND THE SOURCE AND TARGET

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
A1 (bell → paper)	source = bell; target = paper	—	source starts the energy; target changes	naming paper as source	word order
A2 (flashlight → card)	source = flashlight; target = card	—	—	—	—
A3 (warm water → spoon)	source = warm water; target = spoon	—	—	"cold spoon" as source	—
A4 (battery → bulb)	source = battery; target = bulb	—	—	—	—

Section B — WHAT IS THE EVIDENCE?

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
B1	evidence = the change (paper moved /	any observed change	evidence = observable change in target	"energy is the evidence"	informal wording

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
	card lit / temp changed / bulb lit)				
B 2	"We do NOT see energy; we see a change and infer transfer."	"we see the change, not the energy"	change is observed; transfer is inferred	"I saw energy"	—

Section C — TRANSFER / TRANSFORMATION / BOTH

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
Flashlight	BOTH	BOTH (moves + changes form)	energy moves + becomes light	"only transfer"	one-word answer
Buzzer	BOTH	BOTH	moves + becomes sound	"only transformation"	—
Warm water → spoon	TRANSFER	TRANSFER	heat moves; no form change (Grade 4)	"transformation"	—

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
Motor	BOTH	BOTH	moves + becomes motion	"only transfer"	—

Section D — FIX THE EXPLANATION

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
D1 "Energy disappeared when the buzzer made sound."	"Energy did NOT disappear — it transferred AND changed form to sound."	"Energy moved and changed into sound."	energy is not destroyed; it transfers/transforms	"energy disappears / is used up"	informal phrasing

Section E — SHORT EXPLAIN

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
E	Names a source, a change in a target, and	any example with source +	change = evidence → transfer	"energy is directly visible"	spelling; informal

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
	states the change is evidence of transfer (e.g., "The flashlight lit the card; the card lighting up is evidence light transferred energy.")	change + transfer	inferred		wording

Notes

- Accept BOTH for flashlight/buzzer/motor without forcing either/or.
- Warm water → spoon is TRANSFER only at Grade 4 (heat moves; no required form-change framing).
- Reject any claim that energy "disappears" or is "used up."

Answer Key — Chapter 4 Quiz (Energy Transfer)

- **Artifact:** `production/quizzes/04-ch4-energy-transfer-quiz.docx`
- **Source alignment:** Reader Ch.4 pp. 17–24 · TG Lessons 6–9 · AP 6.1–9.2 · PE 4-PS3-2

#	Correct/ expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception	Scoring note
1	energy transfer = energy	"energy	transfer	the Ch.4	transfer = transformation	1 pt; must includ

#	Correct/expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception	Scoring note
	moves from place to place or object to object	moving"	= moves	definition		"move
2	evidence = an observable change in a target	"a change you can see"	change = evidence	evidence is observed, transfer is inferred	"energy itself is the evidence"	1 pt
3	Sound transferred energy (cup + paper; paper moved without contact)	"sound made the paper move"	sound → paper moved	change without contact = transfer	"sound is matter"	1 pt
4	Light transferred energy (flashlight → target lights up)	"light moved to the card"	light → target lights	lit target = change	"I saw energy"	1 pt
5	Heat transferr	"heat moved	warmer →	temperature change =	"cold moved"	1 pt

#	Correct/expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception	Scoring note
	ed energy (warm water cooled, cooler object warmed)	from warm to cool"	cooler	evidence		
6	Electric current transferred energy (battery → device changes)	"battery powered the bulb"	battery → device changes	distant change = transfer	"used up before reaching device"	1 pt
7	Diagram: source → path → target labeled correctly	any correct order	flashlight → beam → card	model orders transfer	wrong direction	2 pts (sourc path/t get)
8	Data: correct reading (warm water ↓, cooler object ↑) with °C	plausible direction	warm ↓ / cooler ↑	transfer warmer → cooler	"both warmer"	2 pts
9	Flashlight =	BOTH	moves + changes	both transfer	"only	1 pt

#	Correct/expected response	Acceptable alternatives	Expected pattern	Reasoning	Common misconception	Scoring note
	BOTH		form	and transformation	transfer"	
10	Buzzer = BOTH	BOTH	moves + becomes sound	both	"only transformation"	1 pt
11	Short explanation: names a change + states it is evidence of transfer (claim "because" + evidence)	any source + change + transfer	change → evidence → transfer	the CER anchor	"I saw energy"	3 pts (claim + evidence + reasoning)

Out-of-scope confirmation

- No compression-wave / electron-flow / collision / engineering / kinetic-potential items.
- No wall-outlet, hot-plate, or coin-cell content.
- Item 9/10 accept BOTH without forcing false either/or.

Total

- 11 items · 15 points (items 1–6, 9, 10 = 1 pt; 7, 8 = 2 pts; 11 = 3 pts).